

Quantum Computational Modeling for Affective Assessment in Virtual Reality Systems

Allison Bayro*
Arizona State University

Heejin Jeong†
Arizona State University

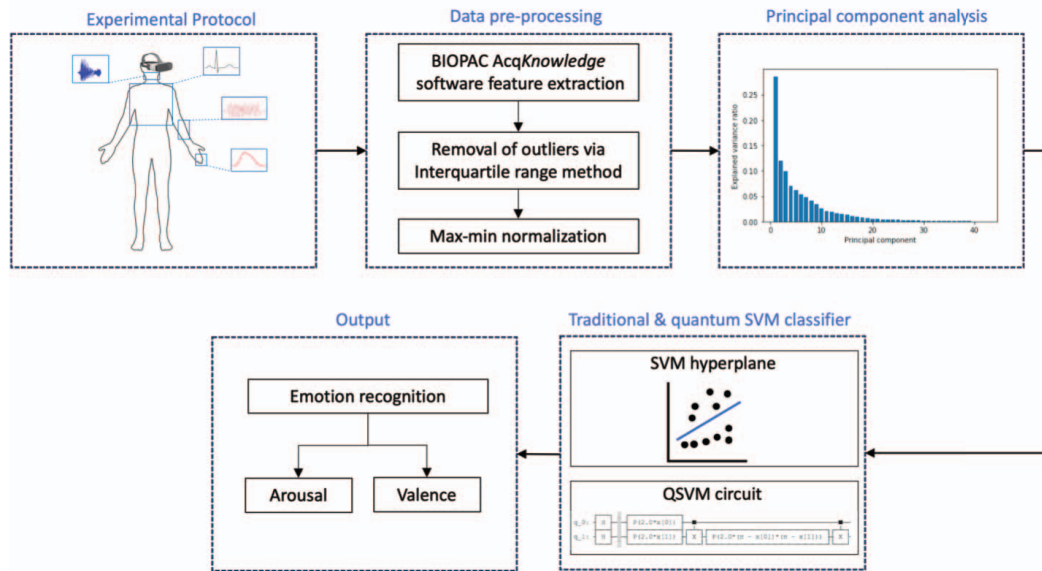


Figure 1: Modeling Approach for Quantum-driven Emotion Recognition.

ABSTRACT

This research explores the use of quantum machine learning—more specifically, the Quantum Support Vector Machine (QSVM)—to predict emotional reactions triggered in virtual reality (VR) settings. The study's experimental setup involved provoking different emotional states via VR gameplay elements and capturing the physiological signals of participants. These signals were paired with the subjects' self-reported emotions, measured through the Self-Assessment Manikin. The study then employed both conventional Support Vector Machines (SVM) and QSVM to classify the data. Notably, QSVM surpassed the traditional SVM in accurately predicting the levels of arousal and valence, achieving higher precision with a reduced set of features. This enhanced efficiency is likely due to QSVM's superior handling of the intricate patterns in emotional data and the quantum models' more effective computational resource usage. These results hold considerable potential for the field of affective computing within VR environments, indicating the advantageous prospects of quantum machine learning for the domain.

Index Terms: Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User models

*e-mail: abayro@asu.edu

†Corresponding author, e-mail: heejin.jeong@asu.edu

1 INTRODUCTION

Affective computing, an interdisciplinary field coined by Picard [14], aims to create systems that can recognize, interpret, process, and simulate human emotions [14]. Its relevance spans various applications, from human-computer interaction and virtual reality (VR) to gaming, mental health, and education, emphasizing the need for emotionally intelligent technology [18]. VR stands out as a significant innovation in human-computer interaction, simulating environments that immerse users and enhance experiences [6]. When combined with affective computing, VR can offer personalized interactions, deepening user engagement and immersion [16].

Machine learning, especially Support Vector Machines (SVMs), has been pivotal in affective computing. SVMs are supervised learning models used for classification and regression, effective in analyzing emotions [4, 7, 8, 19, 20]. However, traditional SVMs often struggle with the complexity and size of affective data, leading to computational challenges [11]. Quantum Support Vector Machines (QSVMs) emerge as a robust alternative, using quantum mechanics to tackle high-dimensional data efficiently [15], as shown in initial explorations by Havlicek et al. [9].

This study delves into the advancements of QSVMs in affective computing, specifically within VR emotion recognition. An overview of this study is presented in Figure 1. We scrutinize both SVMs and QSVMs, assessing their capabilities with complex physiological data, which is crucial for advancing research in this domain. QSVMs harness quantum properties like superposition and entanglement, offering computational advantages over classical SVMs [15]. Subsequent research, including the Quantum Kernel Estimator [13, 21] has furthered QSVMs' potential, showcasing their effectiveness in large-scale machine learning tasks and their capacity to overcome traditional SVMs' limitations. Our work contributes

to this field by offering a comprehensive comparison of SVMs and QSVMs, providing insights to guide future research, and outlining QSVMs' role in the evolution of affective computing.

2 METHODS

2.1 Participants

Our study comprised a diverse group of 30 individuals of 14 females and 16 males, ranging from 18 to 62 years old. The average age was 27 years, with a standard deviation of 8.72.

2.2 Pong Game

We created a tailored virtual environment (VE) using the Unity platform. The foundation of this environment was a modified Air Hockey Game Kit template from the Unity Asset Store, which we customized to support Oculus head-mounted display (HMD) functionality and VR integration [1].

To induce different emotional states, we designed three distinct versions of the VE: fast-paced, slow-paced, and lag-induced. The fast-paced version was engineered for high responsiveness, with an increased frame rate and reduced latency. The game's physics engine was optimized, and visual clutter was minimized to maintain system performance. The gameplay speed was accelerated by tripling the puck's velocity, and the mallet's response to the player's movements was made instantaneous, providing a challenging and engaging opponent at the highest difficulty setting. The slow-paced version aimed for a calmer experience by reducing the puck's speed and the game's overall intensity. The difficulty was set to the lowest, offering a less challenging AI opponent and facilitating a relaxed gaming atmosphere. Finally, the lag-induced version introduced intentional lags and glitches to disrupt the gaming experience. The Unity frame rate was manipulated to create an inconsistent and "laggy" environment, while the game's physics engine and inputs were intermittently frozen, causing unpredictability in the gameplay. This setup challenged players to adapt to and anticipate the erratic behavior of the mallet and puck, allowing us to observe their emotional responses to these disturbances.

2.3 Measurements

The Self-Assessment Manikin (SAM) is a recognized tool for measuring emotional states in affective computing, using visual representations of emotion across dimensions of pleasure-displeasure (valence) and activation-calmness (arousal), allowing participants to rate their feelings on a 9-point Likert scale [5].

We collected physiological data through GSR, ECG, and EMG sensors paired with a BIOPAC Systems data acquisition system, and recorded participants' verbal responses post-gameplay with a MAONO dynamic microphone (MAONO Technology Co.).

For GSR, we prepped the non-dominant hand's fingers with water before applying isotonic gel to the electrodes, which were then placed on the fingers and secured with medical tape. ECG sensors, after cleaning the chest with BIOPAC's skin abrasion gel and an alcohol pad, were positioned according to the Einthoven 3-lead configuration with gel applied for better sensor-skin contact.

Muscle activity for EMG was measured at the flexor digitorum superficialis (FDS) and extensor carpi radialis longus (ECRL), following guidelines by Hermens et al. [10] and Konrad [12]. The sites were prepped with abrasion gel, alcohol, and a conductive gel before attaching two electrodes aligned with each muscle, and a reference on the elbow.

Data analysis and feature extraction for ECG, GSR, and EMG utilized Biopac AcqKnowledge software, following protocols from Bayro et al. [2, 3] for the ECG and GSR. EMG was analyzed with AcqKnowledge's standard functions. Vocal response analysis employed MATLAB, specifically the *audioFeatureExtractor* function, to derive pitch, zero crossing rate, and short-term energy.

2.4 Experimental Design

The main objective of our experiment was to evaluate the emotional reactions and corresponding physiological changes that arise under different gaming conditions. We designed the study with a counter-balanced within-subject methodology.

Participants began their lab visit by signing a consent form and completing a demographic questionnaire. They were then fitted with GSR, ECG, and EMG sensors and underwent a five-minute session to establish baseline physiological data. Next, they took part in a two-minute practice round of VR Pong while seated, using a controller in their dominant hand to become familiar with the virtual mallet controls, under consistent gameplay conditions. In the first experimental session, they played one of three Pong variations—slow-paced, fast-paced, or lag-induced—selected at random for five minutes to minimize motion-related artifacts. Post-game, they verbally responded to open-ended questions within the VR environment, with their responses recorded via a positioned microphone. Afterward, they rated their arousal and valence using the SAM on a 9-point Likert scale and took a two-minute break before proceeding to the subsequent sessions, following the same procedure. At the experiment's end, they filled out an open-ended survey on their experience and emotional responses and stated their gameplay preference. Participants were then debriefed, the biosensors were removed, and they received \$15 compensation for their participation. The entire procedure lasted about 75 minutes.

2.5 Data Analysis

2.5.1 Traditional Support Vector Machine Analysis

Using Python, traditional SVM analysis gauged the precision of arousal and valence predictions, using the binary-labeled SAM results; scores of 4 or below were 0, indicating low, and those above 4 were 1, indicating high. We addressed missing values with k-Nearest Neighbors (kNN) imputation, where a missing value x_m is estimated by averaging or majority vote based on the nearest 'k' data points in the feature space.

The dataset split 80-20 for training and testing ensured both model optimization and impartial performance evaluation. Dimensionality reduction via Principal Component Analysis identified the top 10 variance-explaining features, simplifying computation. SVMs with various kernels—sigmoid, linear, and polynomial—were explored. Model evaluation used 5-fold cross-validation, training on four subsets and validating on one, iteratively, for robust performance assessment.

2.5.2 Quantum Support Vector Machine Analysis

In the quantum machine learning approach with the QSVM, the ZZ feature map was used to translate classical data into quantum states for processing. This map takes a data vector $x = (x_1, x_2, \dots, x_n)$ and applies a sequence of quantum gate operations, including rotations based on each x_i and entangling gates like the controlled-Z. The resulting quantum state is $|\phi(x)\rangle = U_{zz}(x)|0\rangle^n$, where $U_{zz}(x)$ is the transformation by the ZZ feature map and n is the qubit count. We experimented with the top 10 features using different quantum entanglement patterns such as circular, linear, and full, and varied the entanglement layers from 1 to 10 to investigate the feature-entanglement interplay. To evaluate the QSVM's predictive strength, we similarly utilized 5-fold cross-validation, partitioning the data into five sets for a balanced training and validation process.

3 RESULTS

3.1 Arousal Detection Accuracy

The results from the five-fold cross-validation of the traditional SVM analysis show that the linear kernel, incorporating nine features, achieved an optimal accuracy rate of 0.6477, indicating its proficiency in processing high-dimensional data for emotion detection.

In contrast, the QSVM model with circular entanglement surpassed the traditional SVM by attaining a higher accuracy of 0.693, notably achieving this with just the top five features for predicting arousal.

3.2 Valence Detection Accuracy

The comparative study of traditional SVMs and QSVMs for the valence dataset shows distinct results. The traditional SVM with a polynomial kernel, using eight features, reached a peak accuracy of 0.6033. In contrast, the QSVM, which utilizes only the leading five features, outperformed the traditional model with a notable accuracy of 0.637. This indicates the QSVM's effectiveness with circular entanglement in achieving more accurate predictions of emotions using a fewer number of features.

4 DISCUSSION

The results demonstrate the QSVM's superiority in emotional state prediction over the traditional SVM. Remarkably, the QSVM outshone the SVM in accuracy for both arousal and valence emotions, achieving these results with a lesser number of features. This suggests a more efficient model with a lower computational overhead.

These outcomes strongly imply that QSVMs are more adept at capturing complex emotion-related data, thanks to their quantum mechanics-based processing advantages [17]. The use of the ZZ feature map in QSVMs efficiently transforms classical data into quantum states, enhancing computational resource utilization. The QSVM's approach of varying entanglement patterns and iterations offers a substantial benefit. By experimenting with different entanglement forms like circular, linear, and full, and repeating these patterns, the QSVM can explore a broader range of feature combinations, potentially uncovering novel and useful patterns to improve the model's accuracy.

In VR, where immersive and interactive experiences are key, understanding and responding to users' emotions is crucial. The enhanced predictive ability of QSVMs could improve real-time emotion recognition in VR, enabling immediate adjustments to improve the user experience. Additionally, our research indicates that QSVMs could be instrumental in analyzing the emotional impact of various VR elements. Their ability to make accurate predictions with a limited number of features could be invaluable in isolating and examining the emotional effects of specific VR aspects. For therapeutic VR applications, which rely on accurately inducing and monitoring emotions, QSVMs could offer more precise and efficient emotion tracking, potentially improving treatment outcomes [16].

Our study, while promising, has certain limitations. The QSVM's performance was compared to a traditional SVM with only a limited set of kernels and feature maps. Future studies could explore a wider range of quantum feature maps and entanglement strategies with QSVMs. Additionally, the VR environment in our study was limited to a simple Pong game. Applying these findings to more intricate and varied VR scenarios is an area for further research, which should also include different VR applications like educational, therapeutic, or social platforms. This would help ascertain the QSVM's generalizability and utility across various contexts. Our results are encouraging, but they warrant further exploration and validation in subsequent studies.

5 CONCLUSION

Our research highlights the advantages of using QSVM in the realm of affective computing, particularly within VR settings. The QSVM model demonstrated superiority over traditional SVM in terms of accuracy, efficiently predicting emotional states using a reduced number of features. This enhanced performance indicates that QSVMs are not only more effective but also more efficient, making them an invaluable tool for VR research. These results suggest that QSVMs have the potential to revolutionize emotion prediction in VR, offering a more refined and accurate approach. Additionally, the use of

VR as a controlled medium for eliciting emotions presents promising opportunities for the evolution of affective computing. Despite certain limitations, which warrant further investigation, this study marks a significant stride in merging quantum computing with VR for improved emotion detection and prediction. Such developments could greatly enrich user experiences in VR, leading to systems that are more tailored and responsive to individual emotional states.

ACKNOWLEDGMENTS

This work was supported by the National Science Foundation Graduate Research Fellowship Program under Grant No. 026257-001.

REFERENCES

- [1] Air hockey game kit: Systems. Accessed: 2023-01-01.
- [2] A. Bayro, C. Buneo, and H. Jeong. Emotion recognition in virtual reality: Investigating the effect of gameplay variations on affective responses. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 67, pp. 1516–1517. SAGE Publications Sage CA: Los Angeles, CA, 2023.
- [3] A. Bayro, Y. Ghasemi, and H. Jeong. Subjective and objective analyses of collaboration and co-presence in a virtual reality remote environment. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 485–487. IEEE, 2022.
- [4] W. Boucsein et al. Publication recommendations for electrodermal measurements. *Psychophysiology*, 49(8):1017–1034, 2012.
- [5] M. M. Bradley and P. J. Lang. Measuring emotion: The self-assessment manikin and the semantic differential. *Journal of Behavior Therapy and Experimental Psychiatry*, 25(1):49–59, 1994.
- [6] G. C. Burdea and P. Coiffet. *Virtual reality technology*. John Wiley & Sons, 2003.
- [7] R. Calvo and S. D'Mello. Affect detection: An interdisciplinary review of models, methods, and their applications. *IEEE Transactions on Affective Computing*, 1:18–37, 2010.
- [8] C. Cortes and V. Vapnik. Support-vector networks. *Machine Learning*, 20:273–297, 1995.
- [9] V. Havlíček et al. Supervised learning with quantum-enhanced feature spaces. *Nature*, 567:209–212, 2019.
- [10] H. J. Hermens et al. European recommendations for surface electromyography. *Roessingh Research and Development*, 8(2):13–54, 1999.
- [11] C. W. Hsu, C. C. Chang, and C. J. Lin. A practical guide to support vector classification. Technical report, Department of Computer Science, National Taiwan University, 2003.
- [12] P. Konrad. *The ABC of EMG: A Practical Introduction to Kinesiological Electromyography, Version 1.0*. Noraxon INC., 2005.
- [13] S. Lloyd et al. Quantum embeddings for machine learning. *Quantum Science and Technology*, 5(3):034007, 2020.
- [14] R. Picard. *Affective computing*. MIT Press, Cambridge, MA, 1997.
- [15] P. Rebentrost, M. Mohseni, and S. Lloyd. Quantum support vector machine for big data classification. *Physical review letters*, 113(13):130503, 2014.
- [16] G. Riva et al. Affective interactions using virtual reality: the link between presence and emotions. *Cyberpsychology & behavior*, 10(1):45–56, 2007.
- [17] M. Schuld, I. Sinayskiy, and F. Petruccione. Prediction by linear regression on a quantum computer. *Physical Review A*, 94(2):022342, 2017.
- [18] J. Tao and T. Tan. Affective computing: A review. *Affective Computing and Intelligent Interaction*, 2005.
- [19] G. Valenza et al. Revealing real-time emotional responses: a personalized assessment based on heartbeat dynamics. *Scientific reports*, 2:311, 2012.
- [20] X. Zhang et al. Bp4d-spontaneous: a high-resolution spontaneous 3d dynamic facial expression database. *Image and Vision Computing*, 58:1–14, 2017.
- [21] C. Zoufal, A. Lucchi, and S. Woerner. Quantum generative adversarial networks for learning and loading random distributions. *npj Quantum Inf*, 5:103, 2019.